

Question

CuSO_4 , H_5IO_6 , $\text{K}_2\text{S}_2\text{O}_8$, and KOH were mixed in a certain proportion, 200 mL of distilled water was added, heated to boiling for 30 min, cooled, and filtered to obtain a blue-green solution (Reaction 1); an appropriate amount of 10% NaOH solution was added to it, allowed to stand for 48 h, and blue-green crystals precipitated. Partial analysis results of the blue-green crystals are as follows:

(1) X-ray diffraction analysis showed that the crystal is a complex salt with crystal water, the cations are Na^+ and K^+ , and the anion is a mononuclear coordination ion centered on copper, with HIO_6^{4-} as the ligand.

(2) 0.2001 g of the crystal was weighed, 10 mL of hydrochloric acid was added, and heated to dissolve the sample, then diluted with 50 mL of distilled water, buffer solution and PAN indicator were added, and titrated with $0.01973 \text{ mol} \cdot \text{L}^{-1}$ EDTA solution, consuming 11.81 mL.

(3) 0.2059 g of the crystal was weighed, 10 mL 25% KI solution and 10 mL 25% sulfuric acid solution were added (Reaction 2), after full reaction, with starch as an indicator, titrated with $0.1024 \text{ mol} \cdot \text{L}^{-1}$ sodium thiosulfate solution, consuming 42.16 mL.

(4) Thermogravimetric analysis of the crystal showed a weight loss of 25.18%, corresponding to the loss of all crystal water.

Which of the following statements are correct:

1. Cu in the crystal is +5 valence
2. The chemical formula of the crystal contains 12 crystal water molecules
3. Without considering crystal water, the sum of the left and right coefficients of the chemical equation of Reaction 1 is 45
4. Without considering crystal water, the difference between the left and right coefficients of the chemical equation of Reaction 2 is 6 (right minus left)

A. All other options are incorrect

B. 1

C. 2

D. 3

E. 4

F. 1,2

G. 1,3

H. 1,4

I. 2,3

J. 2,4

K. 3,4

L. 1,2,3

M. 1,2,4

N. 1,3,4

O. 2,3,4

P. 1,2,3,4

Answer

Correct Answer: **I**

Detailed Explanation

0.2001 g The copper content in the crystal is: $0.01973 \text{ mol} \cdot \text{L}^{-1} \times 11.81 \text{ mL} = 2.330 \times 10^{-4} \text{ mol}$

Converted to copper content in 1 g crystal: $2.330 \times 10^{-4} \times 1/0.2001 = 1.164 \times 10^{-3} \text{ mol} \cdot \text{g}^{-1}$

CHECKPOINT

1 PTS

1 g crystal contains $1.164 \times 10^{-3} \text{ mol Cu}$

From $\text{I}_2 + 2\text{S}_2\text{O}_3^{2-} = 2\text{I}^- + \text{S}_4\text{O}_6^{2-}$, then $1\text{S}_2\text{O}_3^{2-} \sim 1\text{I}^- \sim 1\text{e}^-$

The amount of electrons transferred by 1 g crystal is:
 $0.1024 \text{ mol} \cdot \text{L}^{-1} \times 42.16 \text{ mL} \times 1/0.2059 = 2.097 \times 10^{-2} \text{ mol} \cdot \text{g}^{-1}$

The number of electrons transferred per mole of crystal reacting with I^- is:
 $(2.097 \times 10^{-2})/(1.164 \times 10^{-3}) = 18.02 \approx 18$

Each mole of HIO_6^{4-} is reduced to I^- transferring 8 mol electrons, so each mole of crystal contains 2 mol ligand HIO_6^{4-}

CHECKPOINT

1 PTS

Each mole of crystal contains 2 mol ligand HIO_6^{4-}

The remaining 2 mol electrons, copper is reduced to +1 oxidation state by I^- , so copper in the crystal is in the +3 oxidation state, statement 1 is incorrect

CHECKPOINT

1 PTS

Copper in the crystal is in the +3 oxidation state

The molar mass of the crystal is: $0.2001 \text{ g} / (2.3301 \times 10^{-4} \text{ mol}) = 858.8 \text{ g} \cdot \text{mol}^{-1}$

The number of crystal water molecules is: $858.8 \text{ g} \cdot \text{mol}^{-1} \times 25.18 \% / 18.02 \text{ g} \cdot \text{mol}^{-1} \approx 12$, statement 2 is correct

CHECKPOINT

1 PTS

The chemical formula of this crystal contains 12 crystal water molecules

Based on the inference of charge conservation, the crystal also contains a total of 5 Na^+ or K^+ , and their total molar mass is:

$$858.8 - 12 \times 18.02 - 63.55 - 2 \times (1.008 + 126.9 + 16.00 \times 6) = 131.2 \text{ (g} \cdot \text{mol}^{-1}\text{)}$$

Which is 1 K^+ and 4 Na^+

The chemical formula is: $\text{Na}_4\text{K}[\text{Cu}(\text{HIO}_6)_2] \cdot 12\text{H}_2\text{O}$

CHECKPOINT

1 PTS

The chemical formula of this crystal is: $\text{Na}_4\text{K}[\text{Cu}(\text{HIO}_6)_2] \cdot 12\text{H}_2\text{O}$

Reaction 1 produced the corresponding potassium salt, and then NaOH was added to replace K^+ to obtain $\text{Na}_4\text{K}[\text{Cu}(\text{HIO}_6)_2] \cdot 12\text{H}_2\text{O}$. Here, $\text{S}_2\text{O}_8^{2-}$ is reduced to SO_4^{2-} , and H_2O is also produced. The equation is: $2\text{CuSO}_4 + 4\text{H}_3\text{IO}_6 + \text{K}_2\text{S}_2\text{O}_8 + 16\text{KOH} = 2\text{K}_5[\text{Cu}(\text{HIO}_6)_2] + 4\text{K}_2\text{SO}_4 + 16\text{H}_2\text{O}$, the sum of the coefficients on the left and right sides is 45, statement 3 is correct

CHECKPOINT

1 PTS

Reaction 1 equation is:
 $2\text{CuSO}_4 + 4\text{H}_3\text{IO}_6 + \text{K}_2\text{S}_2\text{O}_8 + 16\text{KOH} = 2\text{K}_5[\text{Cu}(\text{HIO}_6)_2] + 4\text{K}_2\text{SO}_4 + 16\text{H}_2\text{O}$

Reaction 2 produced CuI , I_2 , H_2O and sulfate. The equation is: $\text{Na}_4\text{K}[\text{Cu}(\text{HIO}_6)_2] + 11\text{H}_2\text{SO}_4 + 17\text{KI} = \text{CuI} + 9\text{I}_2 + 12\text{H}_2\text{O} + 2\text{Na}_2\text{SO}_4 + 9\text{K}_2\text{SO}_4$, the sum of the coefficients on the left side is 29, the sum of the coefficients on the right side is 33, the difference is 4, statement 4 is incorrect

CHECKPOINT

1 PTS

Reaction 2 equation is:
 $\text{Na}_4\text{K}[\text{Cu}(\text{HIO}_6)_2] + 11\text{H}_2\text{SO}_4 + 17\text{KI} = \text{CuI} + 9\text{I}_2 + 12\text{H}_2\text{O} + 2\text{Na}_2\text{SO}_4 + 9\text{K}_2\text{SO}_4$

Statements 2 and 3 are correct, choose I